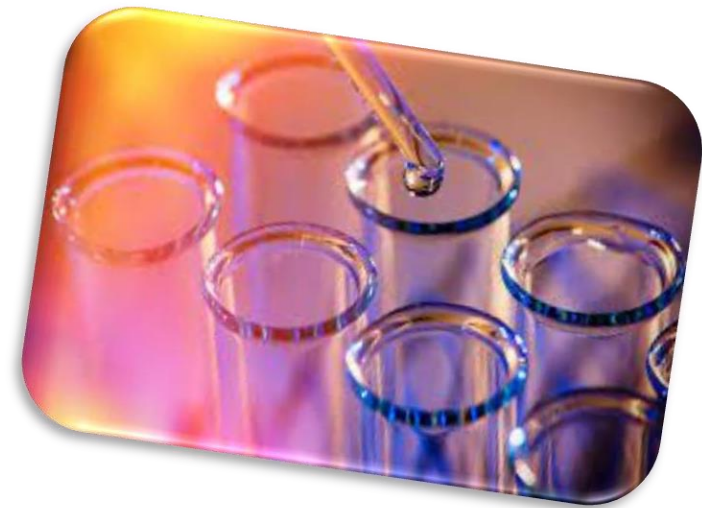
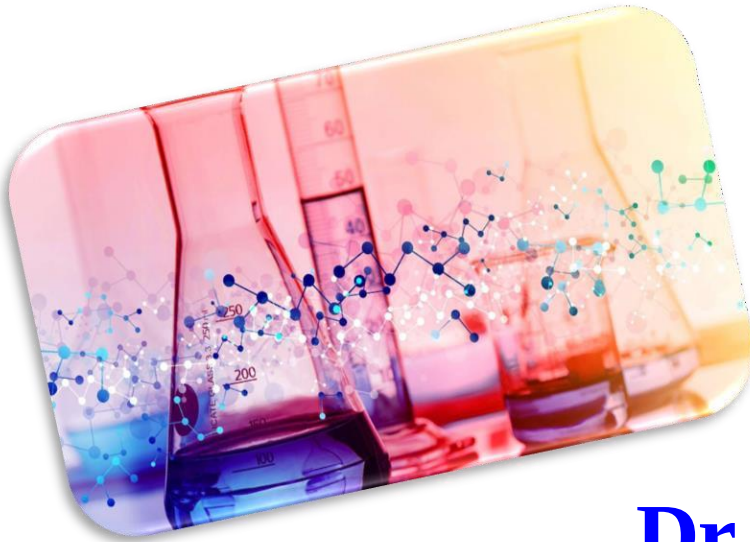




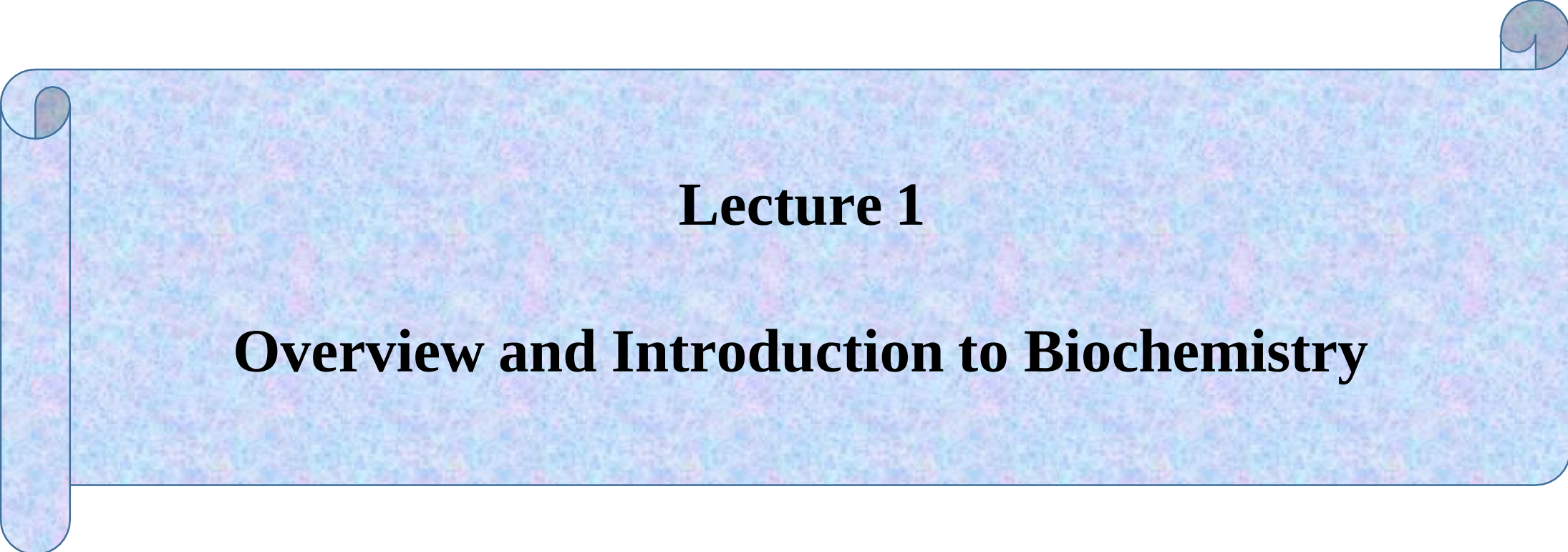
Arab Academy
for Science, Technology & Maritime Transport

Biochemistry EBA1300



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Lecture 1

Overview and Introduction to Biochemistry

Course Overview

- Introduction to biochemistry and carbohydrates
- Basics of cell biology
- Cell transport
- Biochemistry of proteins
- Enzymes
- Genes and gene expression
- Vitamins
- Metabolism

Introduction to Biochemistry

- **Biochemistry**: is the study of the chemical compounds and vital processes occurring in the living organisms.
- Biochemistry means  The chemistry of living organisms.
- The chemical compounds found in the living organisms are called **Biomolecules**.

Main Biomolecules are

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graph TD; A[Main Biomolecules are] --> B[Carbohydrates]; A --> C[Proteins]; A --> D[Lipids]; A --> E["Nucleic acids (DNA, RNA)"]; A --> F[Vitamins]
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Carbohydrates

Proteins

Lipids

**Nucleic acids
(DNA, RNA)**

Vitamins

Carbohydrates (saccharides)

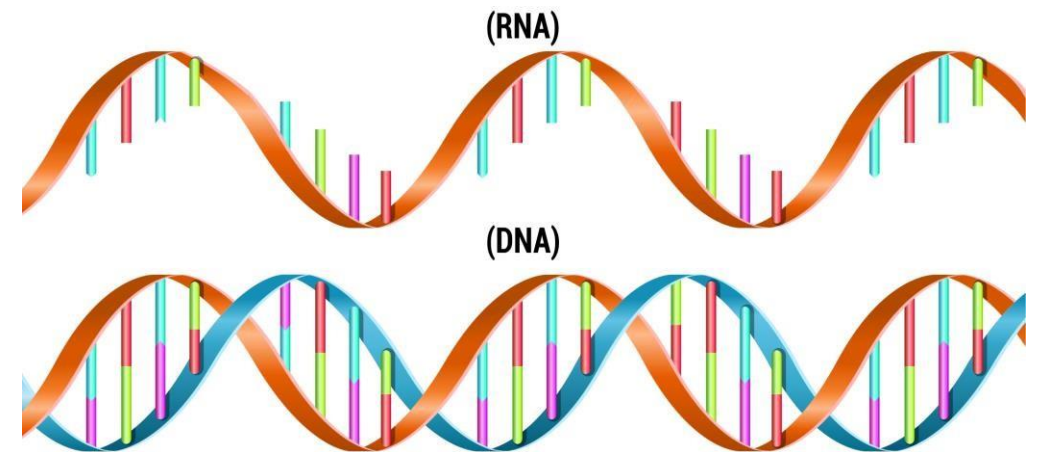


- Carbohydrates are the main nutrients found in foods and drinks.
- Your body breaks down carbohydrates into glucose.
- Some examples of carbohydrate sources are cane sugar, milk, fruits, etc...
- Carbohydrates have a general formula, $C_x (H_2O)_y$, and are called hydrates of carbon from where the name carbohydrate was derived.



Importance of Carbohydrates

- Carbohydrates are essential for life in living organisms, because they are the main source of energy in the body.
- They form a major portion of our food such as fructose, sucrose and lactose.
- Ribose is a part of RNA structure.
- Deoxyribose is a part of DNA structure.

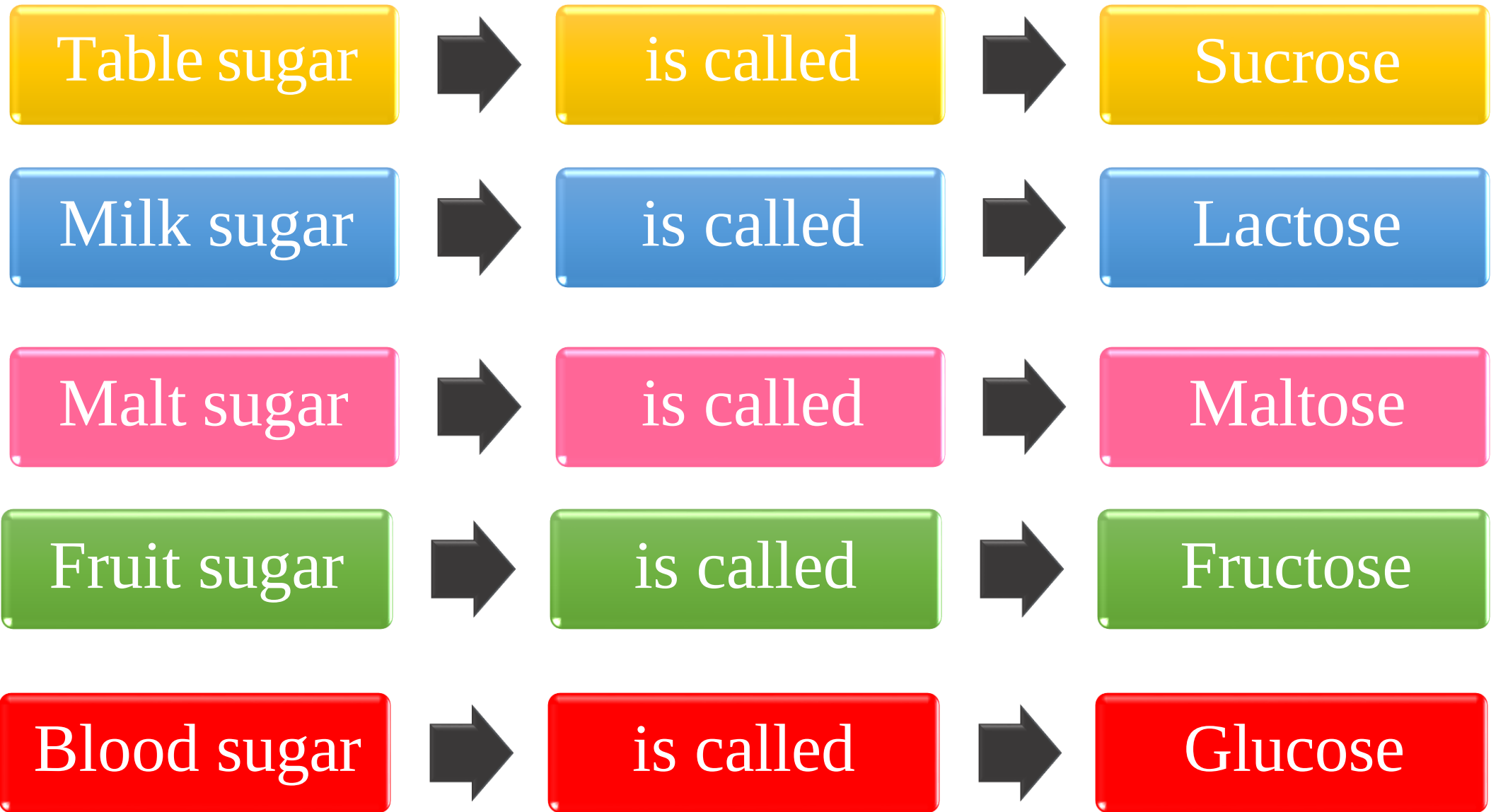


Importance of Carbohydrates

- Carbohydrates are also used as storage molecules (starch in plants and glycogen in animals and human).
- Cell wall of bacteria and plants is made up of cellulose.
- Wood of furniture and cotton of clothes contain cellulose.

Sources of carbohydrates

- Potatoes, wheat, corn, legumes.
- Fruits contain carbohydrates.
- Milk is the only significant source of dietary carbohydrates not derived from plants.



Classification of carbohydrates

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graph LR; A[Classification of carbohydrates] --- B[Monosaccharides<br/>(1 unit of sugar)]; A --- C[Disaccharides<br/>(2 units of sugar)]; A --- D[Polysaccharides<br/>(3 or more units of sugar)];
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Monosaccharides
(1 unit of sugar)

Disaccharides
(2 units of sugar)

Polysaccharides
(3 or more units of sugar)

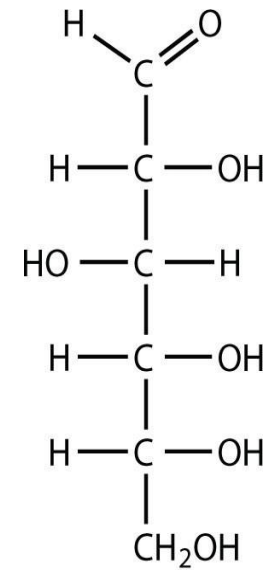
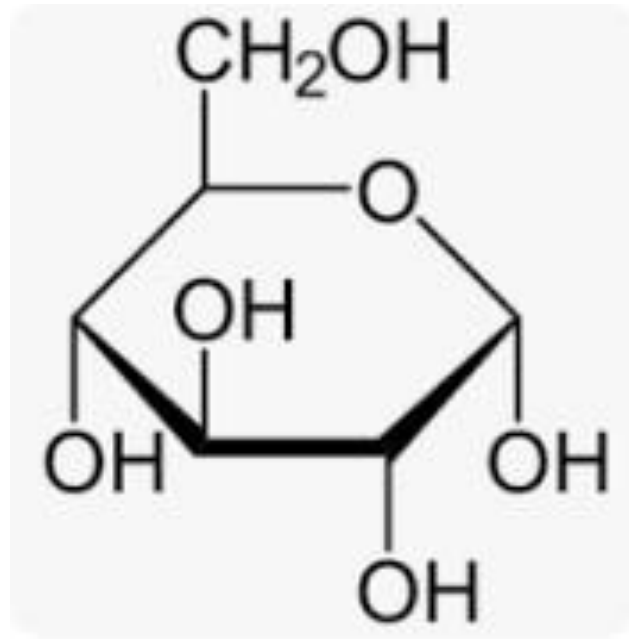
Monosaccharides

- They are the simplest sugars (can't be hydrolyzed or broken to smaller unit).
- Often called simple sugars.
- They are sweeter than disaccharides and polysaccharides.
- They are water soluble.
- According to **number of carbon atoms**, they are classified into:

Triose (3 C) , Tetrose (4 C), pentose (5 C), hexose (6 C) and heptose (7 C).

Examples of monosaccharides

Glucose



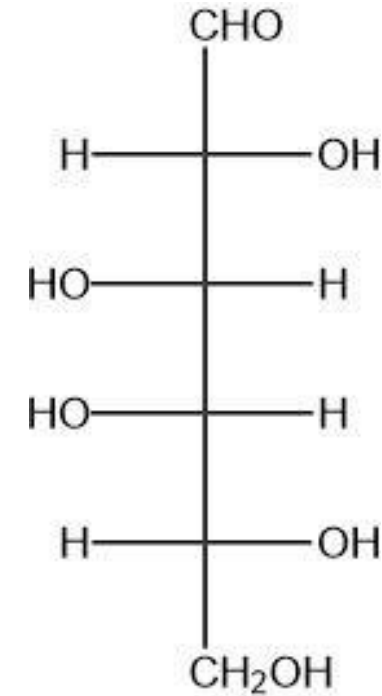
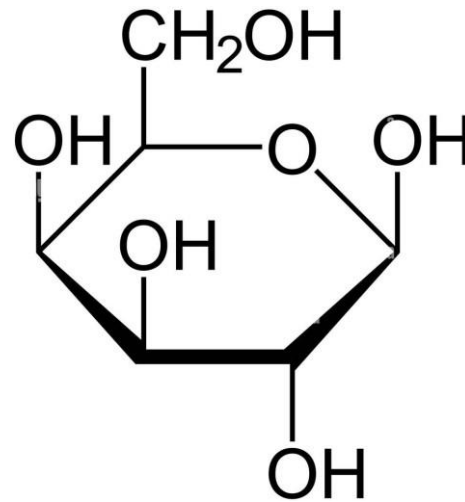
Monosaccharides

Glucose (Blood sugar)

1. The primary fuel for the body.
2. Found in the structure of disaccharides & polysaccharides.

Examples of monosaccharides

Galactose



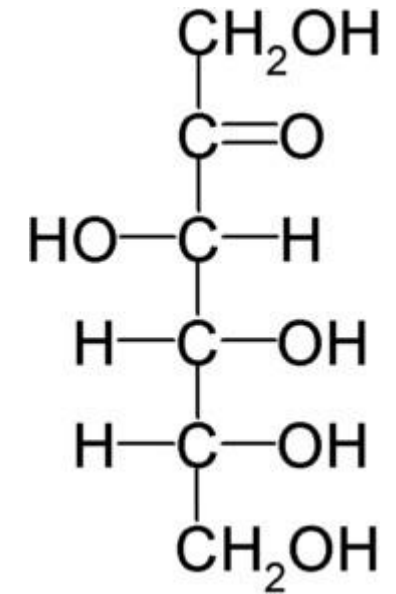
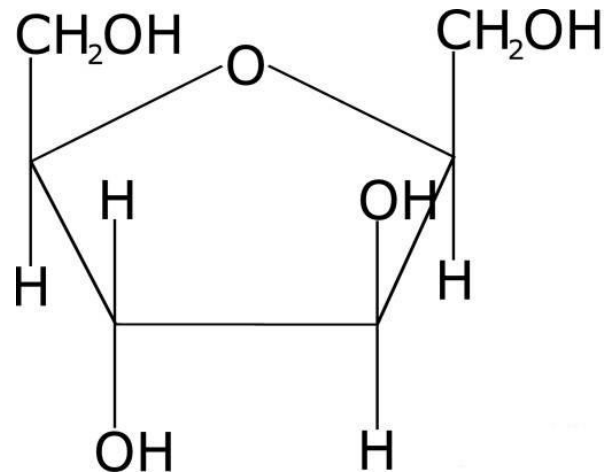
Monosaccharides

Galactose (part of lactose)

1. Found in milk.
2. Converts to *glucose* in the body.

Examples of monosaccharides

Fructose



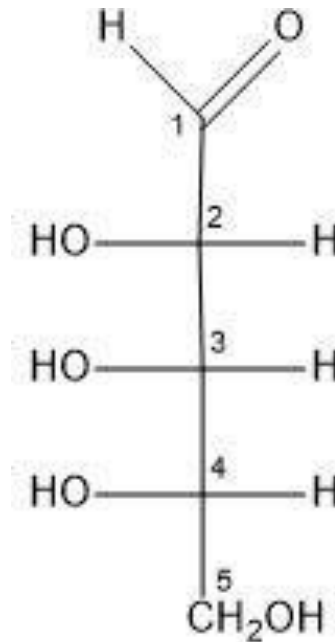
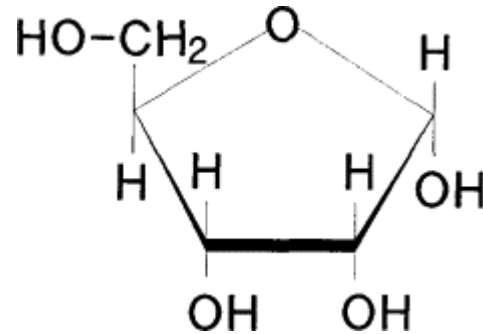
Monosaccharides

Fructose (Fruit sugar)

1. Found in fruit and honey.
2. Converts to *glucose* in the body.

Examples of monosaccharides

Ribose
Nucleic
acid sugar

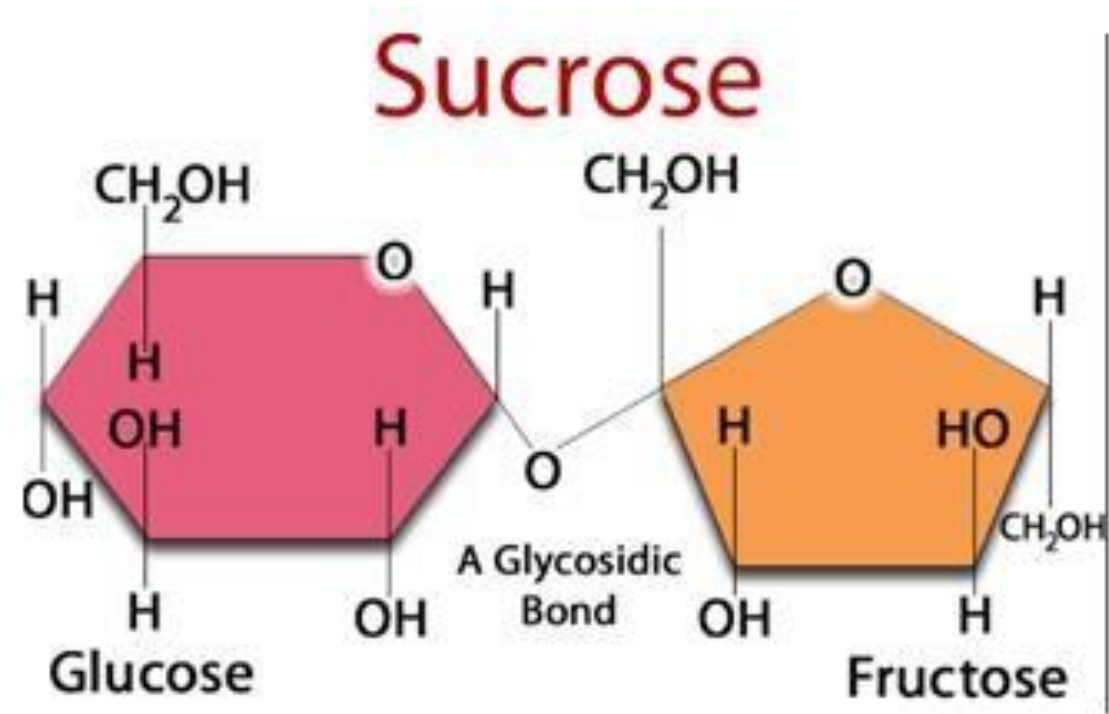


Disaccharides

- These are sugars that contain 2 units of monosaccharides linked with glycosidic linkage.
- **Glycosidic linkage:** Is the linkage between two or more monosaccharide sugars.
- When a disaccharide is hydrolyzed (broken), it gives two monosaccharides.
- Examples of disaccharides are Sucrose (table sugar), Lactose (milk sugar), Maltose (malt sugar).

Examples of Disaccharides

Sucrose



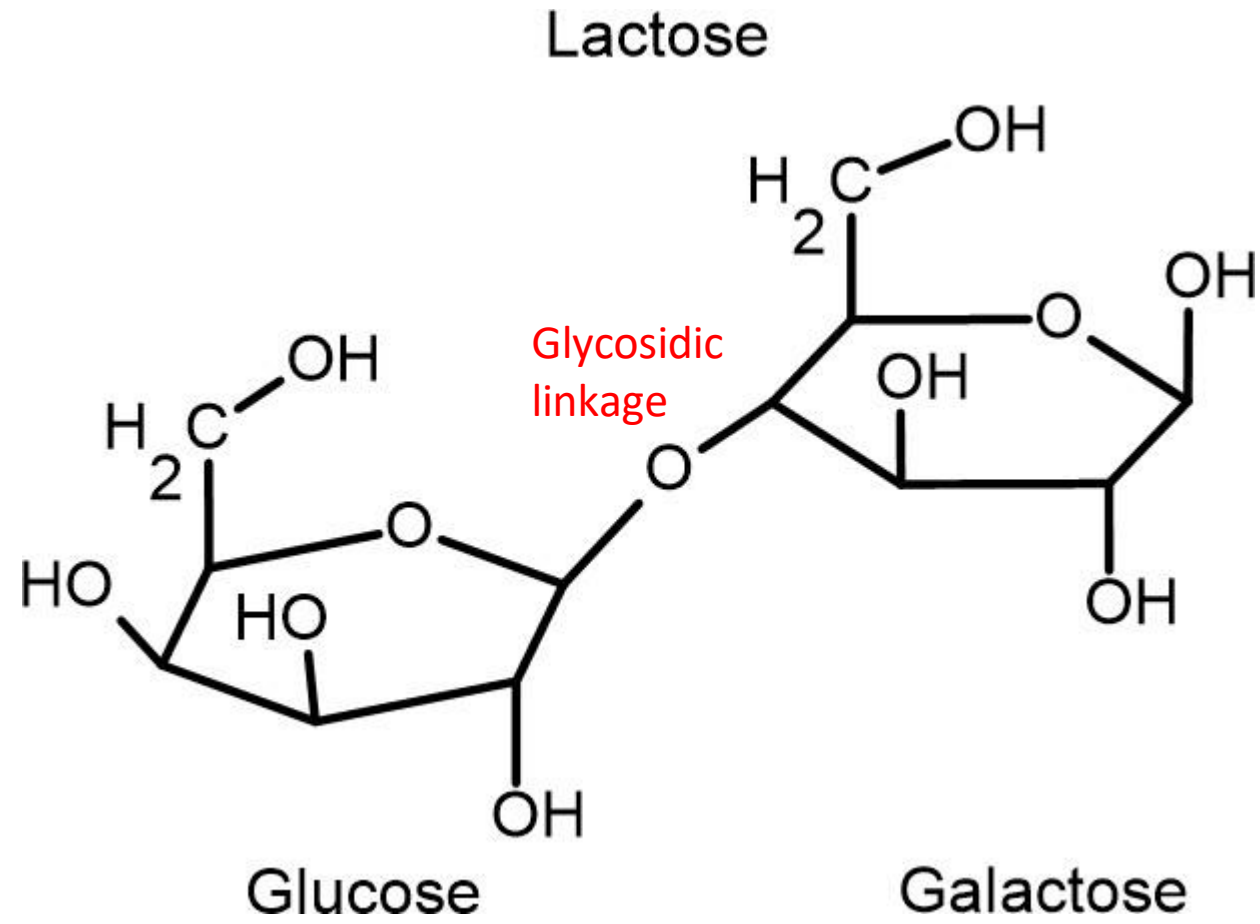
Disaccharides

Sucrose (Table sugar)

1. Glucose + Fructose.
2. Refined from sugar beets & cane.

Examples of Disaccharides

Lactose



Disaccharides

Lactose (Milk sugar)

1. Glucose + Galactose
2. Disease !

Lactose intolerance

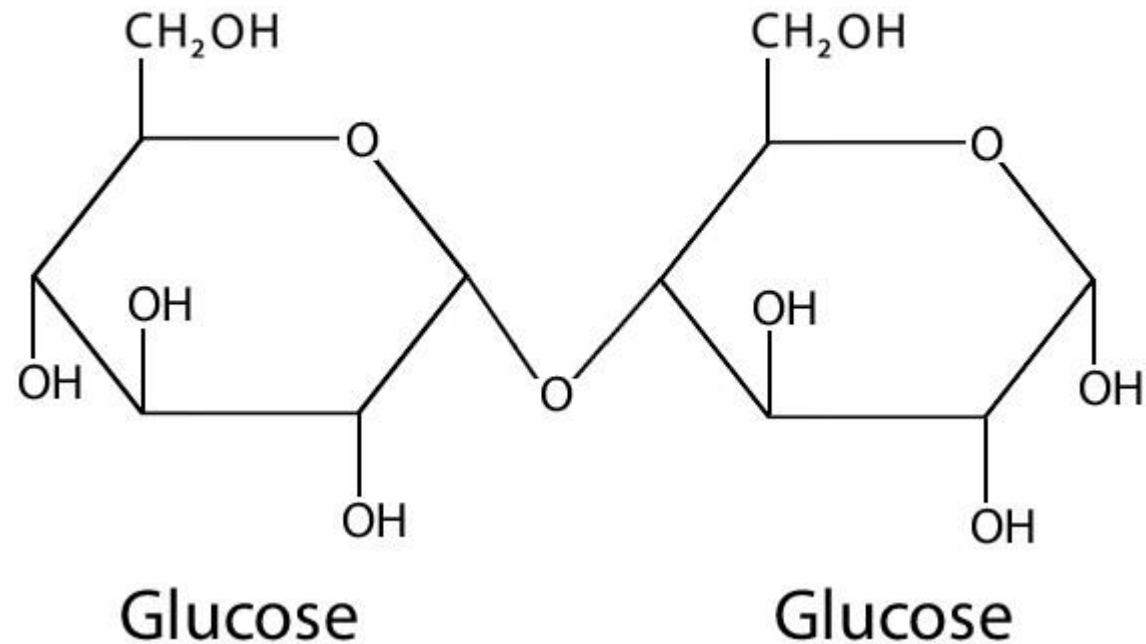


Missing of the digestive enzyme that digest lactose into Glucose + Galactose, so lactose can't be absorbed.

Examples of Disaccharides

Maltose

Maltose



Disaccharides

Maltose (Malt sugar)

1. Glucose + Glucose
2. Found in germinating seeds.

Sucrose



Glucose



Fructose

Lactose



Glucose



Galactose

Maltose



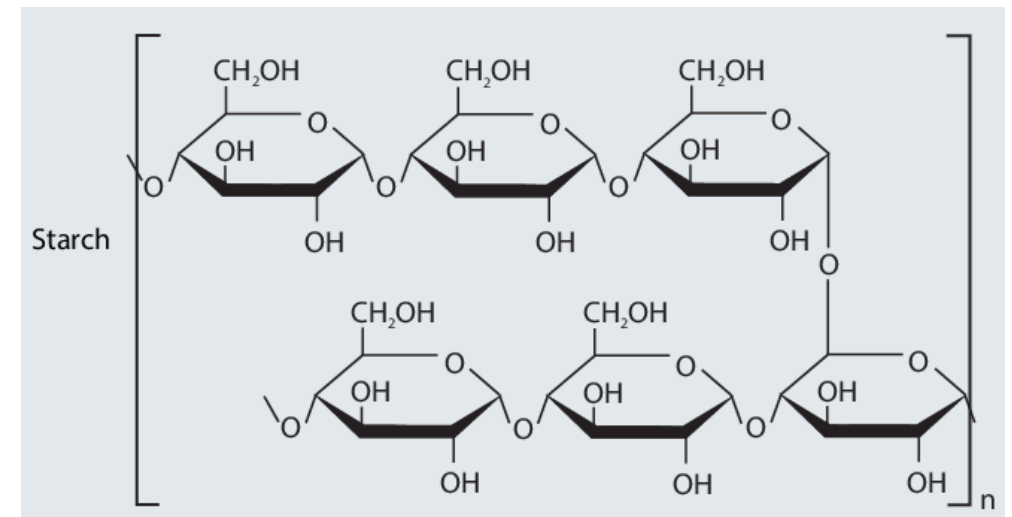
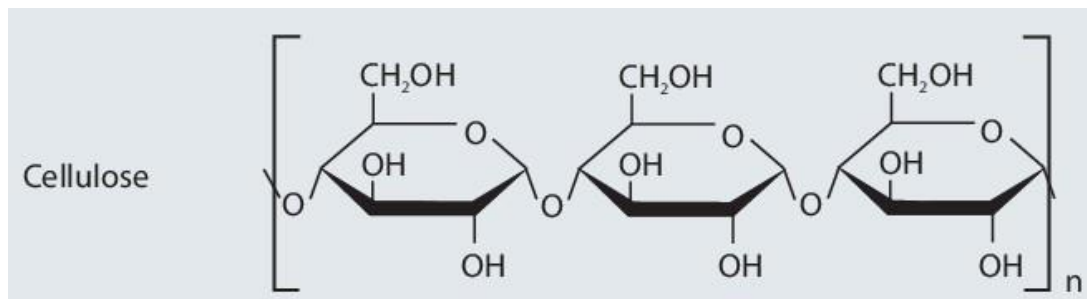
Glucose



Glucose

Polysaccharides

- They are sugars contain 3 or more units of monosaccharides linked by glycosidic linkage.
- All polysaccharides are insoluble in water.
- **Glycogen** is a polymer of glucose that is stored in liver or muscle of human and animals.
- Polysaccharides can be **linear** such as **cellulose** or **branched** such as **starch**.

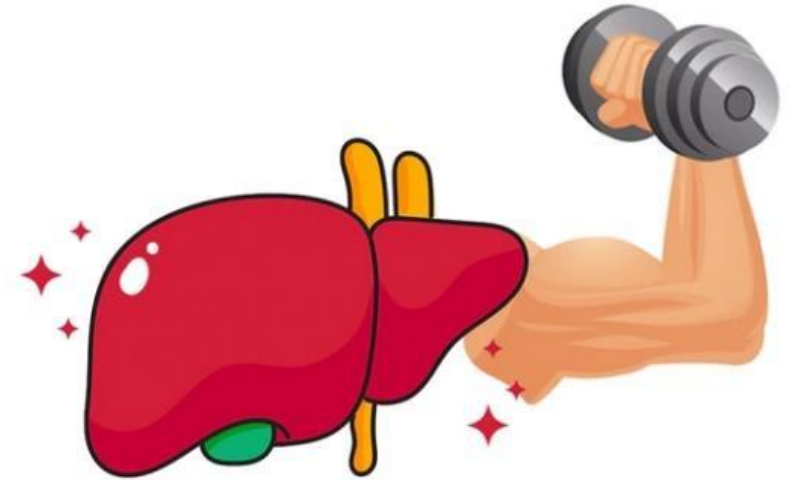


Polysaccharides

Glycogen

(long chains of glucose found in animals and human)

1. Stored in liver & muscles.
2. Helps maintain blood glucose.



Polysaccharides

Starch : long chains of glucose found in wheat, rice, corn, legumes, potatoes.



Monosaccharides

Glucose, Fructose, Galactose, Ribose

Disaccharides

Sucrose, Lactose, Maltose

Polysaccharides

Glycogen, Cellulose, Starch

Questions

Complete the following:

- 1 Maltose is a disaccharide composed of.....and.....monosaccharides.
- 2is the linkage between two or more monosaccharide sugars.
- 3are composed of 3 or more units of sugar.
- 4is a disaccharide composed of glucose and fructose.
- 5is the sugar found in blood.
- 6 are hydrates of carbon .
- 7is a polysaccharide found in liver and muscle.
- 8is a linear polysaccharide.
- 9is a branched polysaccharide.
- 10 Milk sugar is called....., while fruit sugar is called.....



Good Luck!